

Accelerating Low-Frequency Convergence for Limited-Angle DBT via Two-Channel Fidelity in PDHG

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Abstract—Reconstruction in limited-angle digital breast tomosynthesis (DBT) suffers from slow convergence of low spatial-frequency components when using weighted data-fidelity terms within primal-dual optimization. We introduce a two-channel fidelity strategy that decomposes the sinogram residual into complementary low-pass and high-pass bands using square-root Hanning ($\text{Hann}^{1/2}$) filter families, each driven by an independent ℓ_2 -ball constraint and dual update in the PDHG (Chambolle–Pock) algorithm with He–Yuan predictor–corrector relaxation. By assigning a larger dual step-size and slightly looser tolerance to the low-frequency channel, the method delivers stronger per-iteration correction to the near-DC band without violating global PDHG stability. Experiments on a 2D digital breast phantom across multiple resolutions demonstrate that the two-channel approach yields 19%–61% RMSE improvement over the single-channel baseline, with larger gains at coarser discretizations where problem conditioning is more favorable, supporting more balanced spectral convergence in clinically realistic limited-angle regimes.

Index Terms—Computed tomography, Breast tissue, Sparse matrices, Filter banks

I. INTRODUCTION

Limited angular range (LAR) scanning has been widely studied in the literature, but only recently have discrete-to-discrete models demonstrated stable inversion at clinically relevant scan arcs. Building on the observation that directional sparsity improves LAR recovery, Zhang et al. [2021] introduced a globally convex formulation using directional total variation (DTV), enabling robust optimization-based reconstruction.

We introduce a sparsity-constrained optimization framework for LAR DBT that combines directional TV penalties with pixel/voxel sparsity, enabling practical parameterization when true DTV values are unknown. To characterize the regime in which the proposed two-channel fidelity approach is most effective, we systematically vary image discretization to modulate problem conditioning and quantify the resulting reconstruction benefit.

II. PRELIMINARY

We consider limited angular range (LAR) digital breast tomosynthesis (DBT) in a discrete-to-discrete setting. Let f be the discretized attenuation image, g is the measured sinogram

data, and X is the system matrix that encodes fan-beam (2D) or cone-beam (3D) projection. The idealized data model is

$$g = Xf \quad (1)$$

In the DBT geometry, the detector spans the in-plane directions (x, y) , while the z -direction is perpendicular to the (fixed) detector plane and is referred to as the depth direction. At clinically typical scanning arcs, the resulting linear system is generally underdetermined at useful (e.g., isotropic) discretizations, motivating the use of convex regularization.

A. Directional Finite Differences and Directional Total Variation

Let $\partial_x, \partial_y, \partial_z$ denote finite-difference operators approximating derivatives along the corresponding axes. The associated directional total-variation (DTV) seminorms are $\|\partial_x f\|_1$, $\|\partial_y f\|_1$, and $\|\partial_z f\|_1$ (in 2D studies, the ∂_y term is omitted). Distinguishing in-plane and depth directional derivatives is particularly important in LAR/DBT because the data support and conditioning can differ substantially across directions.

B. Constrained Sparsity-regularized Reconstruction

Zhang et al. [2021] showed that accurate image reconstruction from LAR data can be achieved using non-smooth convex optimization with directional total variation (DTV). Their key problem minimizes a least-squares (LSQ) data-fidelity term subject to separate ℓ_1 bounds on the in-plane and depth finite-difference derivatives, forming DTV-constrained LSQ (DTV-LSQ):

$$\begin{aligned} f^* &= \arg \min_f \frac{1}{2} \|g - Xf\|_2^2 \\ \text{s.t. } &\|\partial_x f\|_1 \leq \gamma_x, \|\partial_z f\|_1 \leq \gamma_z. \end{aligned} \quad (2)$$

Constraining DTV is more effective than the standard total variation (TV) constraint based on gradient sparsity. DTV constraints are convenient for phantom studies where true DTV values are computable, but for real digital breast tomosynthesis (DBT) data the bounds must be estimated. The present work reformulates the problem with improved parameterization and design choices to support isotropic-resolution reconstruction in DBT.

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III. THE SINGLE CHANNEL APPROACH

In this section, we describe the single channel approach proposed by Sidky et al. [2025]. Let X be the discrete projection operator for a limited-angle DBT scan, g the measured sinogram, and $f \geq 0$ the unknown volume. The paper adopts a *data-discrepancy constrained, sparsity-regularized* model:

$$\begin{aligned} \min_{f \geq 0} \quad & \alpha_x \|\partial_x f\|_1 + \alpha_y \|\partial_y f\|_1 + \alpha_z \|\partial_z f\|_1 + \beta \|f\|_1 \\ \text{s.t.} \quad & \|R[c](g - Xf)\|_2 \leq \varepsilon \sqrt{\text{size}(g)}, \end{aligned} \quad (3)$$

where $R[c]$ is a square-root Hanning ($\text{Hann}^{1/2}$) filter along detector channels with cutoff parameter c ; $\partial_\bullet$ are finite differences (directional TV). The resulting problem is convex but non-smooth (indicator constraint, ℓ_1 penalties), and is solved with a primal-dual hybrid gradient (PDHG / Chambolle–Pock) method with a He–Yuan predictor–corrector (relaxation) using $\rho \approx 1.75$ He and Yuan [2012]. The limited-angle regime reports running ~ 2000 PD iterations to reach accurate convergence.

There are two additional design choices emphasized in the paper: (i) using $R[c]$ *inside the data-constraint* to de-emphasize low spatial-frequency inconsistencies; (ii) including an optional voxel-sparsity term $\beta \|f\|_1$ in addition to DTV.

a) Notation for PDHG: With $A := R[c]X$ and the residual $r := g - Xf$, the data-constraint is $\|Af - R[c]g\|_2 \leq \varepsilon \sqrt{|g|}$. The PDHG dual variable y lives in the sinogram space; the dual step is a projection onto an ℓ_2 ball; the primal step applies the adjoint $A^\top = X^\top R[c]^\top$ followed by the proximal of the sum of ℓ_1 terms and the indicator of $f \geq 0$.

IV. WHY LOW SPATIAL FREQUENCIES CONVERGE SLOWLY

The data-fidelity term can be written as $\frac{1}{2} \|W(g - Xf)\|_2^2$, where $W \equiv R[c]$. In the detector-frequency domain, this weighting strongly attenuates low spatial frequencies. Under a linearized analysis of the PDHG update, the correction applied to a residual Fourier mode at spatial frequency k scales approximately with the squared singular value s_k^2 of the weighted system operator WX . In limited-angle DBT, the singular values s_k are already small for low k due to the severe ill-conditioning of X . The additional $\text{Hann}^{1/2}$ weighting W further shrinks these low- k singular values by design, creating a large spectral gap between frequency bands.

This disparity in singular values destabilizes convergence behavior. Because PDHG must satisfy the global stability condition $\tau\sigma\|WX\|^2 < 1$ (Theorem 1 of Chambolle and Pock [2011]), the allowable step sizes are dictated by the *largest* singular values, which reside in the mid- and high-frequency bands. Step sizes that stabilize those bands therefore provide insufficient drive for near-DC modes, yielding extremely weak contraction factors for low spatial frequencies. Increasing the global step sizes to accelerate DC correction risks divergence, since PDHG cannot apply frequency-selective step-size amplification in its standard form. Consequently, mid- and high-frequency image components converge quickly, while low spatial frequencies decay slowly and dominate the tail of the iterations. This persistent low-frequency drift reveals

Algorithm 1 PDHG with split fidelity channels

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1: Inputs:  $g, X, R_{\text{hi}}, R_{\text{lo}}, \alpha_{x,y,z}, \beta, \varepsilon_{\text{hi}}, \varepsilon_{\text{lo}}, \sigma_{\text{hi}}, \sigma_{\text{lo}}, \tau, \rho$ 
2: Initialize  $f^0, \bar{f}^0 = f^0, y_{\text{hi}}^0 = y_{\text{lo}}^0 = 0$ 
3: for  $k = 0, 1, 2, \dots$  do
4:    $y_{\text{hi}}^{k+1} \leftarrow \text{Proj}_{B_{\text{hi}}}(y_{\text{hi}}^k + \sigma_{\text{hi}} R_{\text{hi}}(g - X\bar{f}^k))$ 
5:    $y_{\text{lo}}^{k+1} \leftarrow \text{Proj}_{B_{\text{lo}}}(y_{\text{lo}}^k + \sigma_{\text{lo}} R_{\text{lo}}(g - X\bar{f}^k))$ 
6:    $\bar{f}^{k+1} \leftarrow \text{prox}_{\tau(\alpha_x \|\partial_x \cdot\|_1 + \alpha_y \|\partial_y \cdot\|_1 + \alpha_z \|\partial_z \cdot\|_1 + \beta \|\cdot\|_1)}(f^k - \tau X^\top (R_{\text{hi}}^\top y_{\text{hi}}^{k+1} + R_{\text{lo}}^\top y_{\text{lo}}^{k+1}))$ 
7:    $\bar{f}^{k+1} \leftarrow \bar{f}^{k+1} + \rho(\bar{f}^{k+1} - \bar{f}^k) \triangleright \text{He–Yuan relaxation}$ 
8: end for

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a fundamental mode-dependent convergence bottleneck that motivates split-frequency or multi-dual fidelity strategies.

V. TWO-CHANNEL FIDELITY FOR SPECTRALLY BALANCED PDHG

Single-dual weighted fidelity in limited-angle tomographic inversion entangles spatial-frequency bands with sharply different conditioning, which can destabilize convergence when the optimizer attempts to correct near-DC error using a step size dictated by high-frequency modes. To retain the noise robustness of the $\text{Hann}^{1/2}$ weighting while providing a stronger corrective signal to low spatial frequencies, we decompose the sinogram residual into complementary spectral channels and enforce independent consistency constraints:

$$\begin{aligned} \min_{f \geq 0} \quad & \alpha_x \|\partial_x f\|_1 + \alpha_y \|\partial_y f\|_1 + \alpha_z \|\partial_z f\|_1 + \beta \|f\|_1 \\ \text{s.t.} \quad & \begin{cases} \|R_{\text{hi}}(g - Xf)\|_2 \leq \varepsilon_{\text{hi}} \sqrt{|g|}, \\ \|R_{\text{lo}}(g - Xf)\|_2 \leq \varepsilon_{\text{lo}} \sqrt{|g|}. \end{cases} \end{aligned} \quad (4)$$

We select both filters from the square-root Hanning family. The high-frequency channel preserves the original cutoff, $R_{\text{hi}} = \text{Hann}^{1/2}(c)$, maintaining the fidelity weighting used in prior work. The low-frequency channel uses a substantially narrower cutoff, $R_{\text{lo}} = \text{Hann}^{1/2}(c_{\text{lo}})$ with $c_{\text{lo}} \ll c$, forming a complementary pair that partitions spectral correction responsibilities while keeping $R_{\text{hi}}^2 + R_{\text{lo}}^2$ close to identity over the inversion-relevant passband. The low-frequency constraint tolerance is chosen slightly looser than the high-frequency channel to allow mild model mismatch at DC while still driving large-scale bias reduction.

a) Spectral rationale: In primal–dual gradient schemes, the convergence rate of each Fourier mode is governed by the corresponding singular values of the forward operator. In limited-angle systems, high-frequency modes are well represented and produce comparatively large singular values, while low spatial frequencies induce much smaller gains. The original Hann weighting amplifies this gap by further suppressing near-DC singular values. When a single dual is used, the step size must be set to stabilize the high-frequency spectrum, which unintentionally results in an under-driven update for the low-frequency band. The optimizer therefore applies only a weak correction to DC error each iteration, but a strong correction to high frequencies, creating oscillatory or stalled behavior in the low band.

By assigning each channel an independent dual step size, the method reduces cross-band interference. The low-frequency dual ascent can safely be made larger, providing a stronger contraction factor for small k , without compromising the stability required to converge the high-frequency channel. The proximal update then combines both adjoint-filtered corrections while preserving non-negativity. This split-fidelity design yields a more spectrally uniform correction profile and mitigates frequency-induced convergence bottlenecks in limited-angle tomographic inversion.

VI. EXPERIMENTAL METHODOLOGY

To evaluate the proposed two-channel fidelity approach, we conduct inverse problem studies using a 2D LAR scanning configuration matching that of Sidky et al. [2025]. The geometry uses 25 fan-beam projections over a 50° arc. The source is positioned 50 cm from the isocenter and 100 cm from the detector, which consists of 1024 bins spanning a length that inscribes a $10 \text{ cm} \times 10 \text{ cm}$ field of view.

A. Digital Phantom

The phantom used for these studies is a 2D digital breast slice following the methodology of Sidky et al. [2025]. Glandular tissue regions are created by applying a threshold to power-law noise, with small bright specks added to simulate microcalcifications Reiser and Nishikawa [2010]. The test image is constructed as a weighted combination of three tissue components: adipose ($0.5\times$), fibroglandular ($1.0\times$), and calcification ($2.0\times$) layers.

To characterize how the two-channel fidelity benefit depends on problem conditioning, we test three array sizes: 512×512 pixels (0.2 mm pixel width), 256×256 pixels (0.4 mm pixel width), and 128×128 pixels (0.8 mm pixel width). Because the LAR inverse problem has a fixed number of measurements ($25 \text{ views} \times 1024 \text{ detector bins} = 25,600 \text{ ray integrals}$), coarser discretizations yield fewer unknowns and thus better-conditioned systems. This experimental design allows us to isolate the relationship between conditioning and the efficacy of the split-frequency fidelity strategy. Following Sidky et al. [2025], projection data are generated using the same pixel grid as the reconstruction, which removes discretization error and focuses the comparison on algorithmic differences. All studies use noiseless projection data to isolate the convergence behavior of the optimization algorithms from noise-related effects.

B. Algorithm Configuration

Both reconstruction methods—single-channel and two-channel—are configured with matched regularization parameters to ensure fair comparison. Following the resolution-dependent parameterization of Sidky et al. [2025], the DTV weight parameter is set to $\alpha = 1.7$ for 512×512 , $\alpha = 1.9$ for 256×256 , and $\alpha = 1.95$ for 128×128 . The pixel sparsity parameter is set to $\beta = 5.0$ for the 512×512 case and $\beta = 10.0$ for the 256×256 and 128×128 cases.

The PDHG algorithm is run for 500 iterations using the He–Yuan predictor-corrector scheme He and Yuan [2012]

with relaxation parameter $\rho = 1.75$. For the single-channel baseline, the data-discrepancy constraint employs a square-root Hanning filter with cutoff parameter $c = 4.0$. The two-channel method uses complementary filters: a low-pass channel with cutoff $c_{\text{lo}} = 8.0$ and a high-pass channel with cutoff $c_{\text{hi}} = 4.0$. The low-frequency channel employs an amplified dual step size with $\sigma_{\text{lo}} = 4.0 \times \sigma_{\text{hi}}$ and a relaxed tolerance $\varepsilon_{\text{lo}} = 1.25 \times \varepsilon_{\text{hi}}$, enabling stronger per-iteration correction to near-DC components while maintaining global PDHG stability Chambolle and Pock [2011]. Step sizes are determined via power iteration to satisfy the convergence condition $\tau\sigma\|WX\|^2 < 1$. Our implementation is publicly available at <https://github.com/taroi/ct>.

VII. RESULTS

We now present quantitative and visual comparisons of the single-channel and two-channel methods. Table I summarizes reconstruction error across three image resolutions.

TABLE I
IMAGE RMSE COMPARISON ACROSS RESOLUTIONS

Resolution	Single-Channel	Two-Channel	Improvement
512×512	0.1625	0.1317	19.0%
256×256	0.0984	0.0767	22.1%
128×128	0.0332	0.0128	61.4%

The two-channel method achieves consistently lower reconstruction error across all tested resolutions. Notably, the improvement magnitude increases substantially with coarser discretization: from 19.0% at 512×512 to 61.4% at 128×128 . This trend directly reflects the relationship between problem conditioning and the efficacy of split-frequency fidelity. At coarser resolutions, the ratio of measurements to unknowns improves (25,600 measurements versus 16,384 unknowns at 128×128 , compared to 262,144 unknowns at 512×512), yielding a better-conditioned inverse problem. Under improved conditioning, the low-frequency error components—which the two-channel approach specifically targets—become the dominant source of residual error, and the amplified dual step size for the low-frequency channel can provide proportionally greater correction. Conversely, at finer resolutions where severe ill-conditioning introduces errors across all frequency bands, the targeted low-frequency correction constitutes a smaller fraction of the total improvement opportunity.

Figure 1 shows the convergence behavior for the 256×256 case. Both methods exhibit decrease in image RMSE, but the two-channel approach achieves lower error throughout the iteration process. The single-channel method displays pronounced oscillations in the early iterations that persist until approximately iteration 200, whereas the two-channel method stabilizes more rapidly after an initial transient. This difference is consistent with the theoretical motivation: by assigning an independent dual step size to the low-frequency channel, the two-channel approach reduces cross-band interference that would otherwise cause the optimizer to oscillate as it attempts to balance corrections across frequency bands with disparate conditioning.

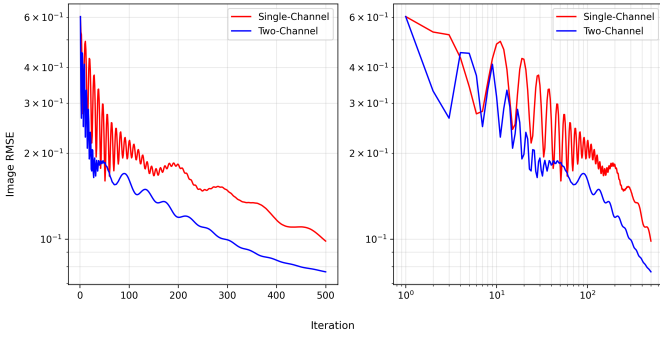


Fig. 1. Image RMSE convergence for 256×256 reconstruction. Left: linear iteration scale. Right: log-log scale. The two-channel method (blue) achieves lower error and reduced early-iteration oscillation compared to the single-channel baseline (red).

Figure 2 presents the reconstructed images and difference maps for the 256×256 case at 500 iterations. Both methods recover the gross phantom structure, including the glandular tissue distribution and microcalcification specks. The difference images reveal that the single-channel method retains more residual error distributed throughout the reconstruction, particularly in the form of large-scale intensity variations characteristic of unconverged low-frequency components. The two-channel reconstruction exhibits more uniform error reduction, with the residual errors appearing more localized to fine structural boundaries.

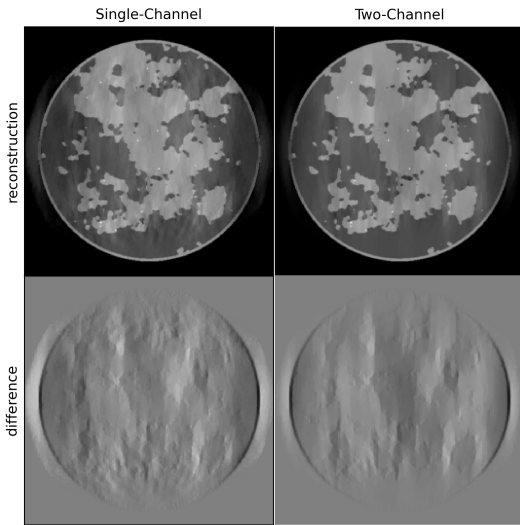


Fig. 2. Reconstruction comparison for 256×256 images. Top row: single-channel (left) and two-channel (right) reconstructions. Bottom row: corresponding difference images (reconstruction minus ground truth), displayed with grayscale range $[-0.75, 0.75]$. The two-channel result shows reduced large-scale intensity artifacts.

These results are consistent with the spectral analysis of Section 4: frequency-separated data fidelity enables more balanced convergence across the spatial frequency spectrum. The benefit is most pronounced when the underlying inverse problem is better conditioned, as occurs at coarser discretizations where the number of unknowns more closely approaches the number

of measurements.

VIII. CONCLUSION

We analyzed the fundamental causes of slow and unstable convergence in limited-angle digital breast tomosynthesis reconstruction when using a single, globally weighted fidelity term in primal-dual optimization. The square-root Hanning weighting creates a large singular-value gap between high and near-DC modes. Because PDHG step sizes must stabilize the largest gains (high frequencies), the low-frequency channel is under-driven, leading to stalled or unstable convergence at DC.

To address this, we proposed a two-channel fidelity design that routes low and high spatial-frequency residuals through complementary square-root Hanning filters, each paired with an independent dual variable and step size. This strategy preserves noise robustness while reducing cross-band step-size interference, providing stronger corrective feedback to the ill-conditioned low-frequency channel without violating global PDHG stability. The resulting formulation better equalizes spectral corrections, mitigates low-band oscillations, and directly targets the most ill-conditioned modes in limited-angle inversion.

Experiments across multiple image resolutions reveal that the two-channel benefit scales with problem conditioning: improvements range from 19% at 512×512 (severely underdetermined) to 61% at 128×128 (moderately underdetermined). This conditioning dependence suggests the method is most impactful when low-frequency errors dominate the residual, as occurs when the inverse problem is sufficiently well-posed for high-frequency components to converge reliably.

These findings support the broader conclusion that accounting for spectral conditioning gaps in reconstruction optimizers for limited-angle tomography can help avoid unstable convergence tails. As future work, we believe that multi-dual or frequency-split fidelity designs can offer a principled and practical path to improving stability and convergence uniformity in other types of tomographic reconstruction.

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